MATH 55 - HW 1

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Problems from Hutchings

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The **elements** of the set $\{\{1,\{2\}\},\{3\}\}$ are:

- $\{1,\{2\}\}$
- $\{3\}$

The **subsets** of the set $\{\{1,\{2\}\},\{3\}\}$ are:

- $\emptyset$
- $\{\{1,\{2\}\}\}$
- $\{\{3\}\}$
- $\{\{1,\{2\}\},\{3\}\}$

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a) $\{\{\emptyset\}\} \cup \emptyset = \{\emptyset, \{\emptyset\}\}$
False. The union of any set with the empty set ($\emptyset$) results in the original set. Therefore, $\{\{\emptyset\}\} \cup \emptyset$ is simply $\{\{\emptyset\}\}$. The set $\{\{\emptyset\}\}$ has only one element, which is $\{\emptyset\}$, while the set $\{\emptyset, \{\emptyset\}\}$ has two elements, $\emptyset$ and $\{\emptyset\}$.

b) $\{\{\emptyset\}\} \cup \{\emptyset\} = \{\emptyset, \{\emptyset\}\}$
True. The union of two sets combines all the distinct elements from both sets. The first set contains the element $\{\emptyset\}$, and the second set contains the element $\emptyset$. Combining them results in the set $\{\emptyset, \{\emptyset\}\}$.

c) $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\emptyset\}$
False. The intersection of two sets contains only the elements that are present in *both* sets.

- The elements of the first set are $\emptyset$ and $\{\emptyset\}$.
- The elements of the second set are $\{\emptyset\}$ and $\{\{\emptyset\}\}$.
- The only common element is $\{\emptyset\}$.
- Therefore, the intersection is $\{\{\emptyset\}\}$, not $\{\emptyset\}$.

d) $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\{\emptyset\}\}$
True. As explained above, the only element common to both sets is
$\emptyset$. The intersection is therefore the set containing only that element, which is $\{\{\emptyset\}\}$.

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Consider an arbitrary element x:

- Suppose $x \in A \cap (B - C)$. Then $x \in A$ and $x \in B - C$, so $x \in A$, $x \in B$, and $x \notin C$. Thus, $x \in A \cap B$ and $x \notin A \cap C$ (since $x \notin C$), so $x \in (A \cap B) - (A \cap C)$.
- Suppose $x \in (A \cap B) - (A \cap C)$. Then $x \in A \cap B$ and $x \notin A \cap C$, so $x \in A$, $x \in B$, and it is not the case that both $x \in A$ and $x \in C$. Since $x \in A$, this implies $x \notin C$. Thus, $x \in A$ and $x \in B - C$, so $x \in A \cap (B - C)$.

The two sets are equal since they contain exactly the same elements.
It is **not always true** that $A \cup (B - C) = (A \cup B) - (A \cup C)$. Consider an arbitrary element x:

- The left side consists of all x such that $x \in A$ or ($x \in B$ and $x \notin C$).
- The right side consists of all x such that ($x \in A$ or $x \in B$) and not ($x \in A$ or $x \in C$). The negation simplifies to $x \notin A$ and $x \notin C$, so the right side consists of all x such that ($x \in A$ or $x \in B$) and $x \notin A$ and $x \notin C$. This further simplifies to $x \in B$ and $x \notin A$ and $x \notin C$.

These are not equivalent in general. For a counterexample, let $A = \{1\}$, $B = \{2\}$, and $C = \{3\}$. Then the left side is $\{1\} \cup (\{2\} - \{3\}) = \{1\} \cup \{2\} = \{1, 2\}$, while the right side is $(\{1\} \cup \{2\}) - (\{1\} \cup \{3\}) = \{1, 2\} - \{1, 3\} = \{2\}$. The sets are not equal.

Problems from Rosen

Section 2.1

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- $A = \emptyset$
- $B = \{\emptyset\}$

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Assume $\mathcal{P}(A) = \mathcal{P}(B)$.

Let $x \in A$. Then $\{x\} \subseteq A$, so $\{x\} \in \mathcal{P}(A)$. Since $\mathcal{P}(A) = \mathcal{P}(B)$, it follows that $\{x\} \in \mathcal{P}(B)$, so $\{x\} \subseteq B$, and hence $x \in B$. Thus, $A \subseteq B$.

Similarly, let $y \in B$. Then $\{y\} \subseteq B$, so $\{y\} \in \mathcal{P}(B)$. Since $\mathcal{P}(B) = \mathcal{P}(A)$, it follows that $\{y\} \in \mathcal{P}(A)$, so $\{y\} \subseteq A$, and hence $y \in A$. Thus, $B \subseteq A$.

Therefore, $A = B$.

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Let $(x, y) \in A \times B$. By definition of the Cartesian product:

- $x \in A$
- $y \in B$

Since $A \subseteq C$: $x \in C$. Since $B \subseteq D$: $y \in D$.
From above, $x \in C$ and $y \in D$, so $(x, y) \in C \times D$.
Since every element of $A \times B$ is also in $C \times D$, it follows that:

$$A \times B \subseteq C \times D$$

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If $A \times B = \emptyset$, then there are no ordered pairs (a, b) such that $a \in A$ and $b \in B$. This means that it is impossible to pick an element from A and an element from B simultaneously. If both A and B were nonempty, then there would exist at least one $a \in A$ and one $b \in B$, so $(a, b) \in A \times B$, contradicting $A \times B = \emptyset$. Therefore, at least one of the sets A or B must be empty, i.e., $A = \emptyset$ or $B = \emptyset$.

Section 2.2

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$$A = (A - B) \cup (A \cap B) = \{1, 5, 7, 8\} \cup \{3, 6, 9\} = \{1, 3, 5, 6, 7, 8, 9\}$$
$$B = (B - A) \cup (A \cap B) = \{2, 10\} \cup \{3, 6, 9\} = \{2, 3, 6, 9, 10\}$$

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No for a).
A counterexample is $A = \{1\}$, $B = \{1, 2\}$, and $C = \{2\}$. Here, $A \cup C = \{1, 2\} = B \cup C$, but $A \neq B$.
No for b).
A counterexample is $A = \{1, 2\}$, $B = \{1, 3\}$, and $C = \{1\}$. Here, $A \cap C = \{1\} = B \cap C$, but $A \neq B$.
Yes for c).
To prove this, assume $A \cup C = B \cup C$ and $A \cap C = B \cap C$.
Let $x \in A$. If $x \in C$, then $x \in A \cap C = B \cap C$, so $x \in B$. If $x \notin C$, then since $x \in A \cup C = B \cup C$ and $x \notin C$, it follows that $x \in B$.
Thus, $A = B$.

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a) Since $A_1 \subset A_2 \subset \cdots \subset A_n$, any element in some A_i (with $i \leq n$) satisfies $k \leq i \leq n$, so it is in A_n. Conversely, every element in A_n is in the union. Therefore, the union $\bigcup_{i=1}^{n} A_i$ is the largest set in the collection, which is A_n. Thus, $\bigcup_{i=1}^{n} A_i = \{k \in \mathbb{Z} \mid k \leq n\}$.

b) Since $A_1 \subset A_2 \subset \cdots \subset A_n$, an element k is in the intersection if and only if $k \leq i$ for every $i = 1, \ldots, n$. The most restrictive condition is $k \leq 1$, and the larger upper bounds are automatically satisfied if this holds. Therefore, the intersection $\bigcap_{i=1}^{n} A_i$ is the smallest set in the collection, which is A_1. Thus, $\bigcap_{i=1}^{n} A_i = \{k \in \mathbb{Z} \mid k \leq 1\}$.